Enhancing Remote Sensing Scene Classification With Hy-MSDA: A Hybrid CNN–Transformer for Multisource Domain Adaptation

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***Abstract*— Multisource unsupervised domain adaptation (MUDA) has demonstrated its effectiveness in improving model performance for remote sensing (RS) scene classification, particularly in cases where the target-domain lacks labeled data. However, most current methods based on convolutional neural networks (CNNs) or Transformers fail to fully exploit the features within each source domain, which benefits classification accuracy. Additionally, these methods frequently overlook the varying levels of similarity between domains, limiting the potential of MUDA. To alleviate this limitation, we propose Hy-MSDA, a hybrid CNN–Transformer with consistent learning and dynamic weighting for MUDA, which fully explores and utilizes valuable information from multiple sources at both the feature and decision levels. To achieve a better alignment of categories among domains, the consistency learning module aims to learn domain-invariant features, maintaining consistency in global high-level features and rich semantic information at the feature level. At the decision level, the dynamic weighting strategy balances the contribution of each source domain by considering their varying importance levels. These two components are closely interconnected and mutually reinforce each other, resulting in improved cross-domain collaboration of Hy-MSDA for domain adaptation. Experimental results on the RS classification datasets clearly demonstrate that Hy-MSDA outperforms the state-of-the-art methods, achieving a significant enhancement of 2%–7% in classification accuracy. Moreover, experiments on RS segmentation datasets reveal that Hy-MSDA performs comparably to supervised methods, further validating the effectiveness of Hy-MSDA in segmentation tasks. The source code is available at https://github.com/phaeton2017/Hy-MSDA.**

***Index Terms*— Consistency learning, dynamic weighting, multisource unsupervised domain adaptation (MUDA), remote sensing, scene classification.**

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I. Introduction

RECENTLY, remote sensing (RS) image scene classification, has advanced significantly in terms of extracting valuable geographic information [1], [2]. However, the diversity in imagery acquisition processes, such as satellite sensors, geographic locations, and geometric resolutions, introduces domain discrepancies. These discrepancies hinder the effective transfer of models from one domain (source domain) to another domain (target domain), resulting in degraded performance. Traditional supervised methods rely heavily on abundant labeled data for training [3], ultimately making them ineffective when labeled data are limited. Moreover, remote sensing images typically cover such large areas that manual annotation is both difficult and costly, limiting the effectiveness of supervised learning methods.

Unsupervised domain adaptation (UDA) has emerged as a viable solution for addressing the above problems [4]. UDA methods can efficiently handle complex backgrounds and diverse land-cover types by uncovering inherent data patterns, thus improving RS scene classification accuracy in scenarios where manual labeling is impractical. However, the effectiveness of UDA methods in enhancing the scene classification performance of the target domain is largely determined by the level of alignment between the source domain and target domains. In fact, there are multiple labeled source domains in RS imagery, making it challenging to determine which source domain best fits the target domain [5].

The data from different domains in RS imagery may contain complementary information. Therefore, transferring multiple source domains' knowledge to the target domain could produce superior results compared with using one source domain alone [5], [6]. However, the similarity between the sample distributions of different domains varies, resulting in varied contributions to the domain adaptation task on the target domain. To effectively utilize valuable information from various source domains, numerous methods in multisource UDA (MUDA) analyze the similarity between the data distributions of domains to evaluate and account for the significance of different source domains [7], [8], [9]. However, it is insufficient to consider only the information from multiple source domains at the data level. These methods fail to account for the model's ability to learn from features, which is a crucial factor in representing domain-invariant features.

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To enhance domain adaptation performance during model training, it is essential to facilitate the ability to learn domain-invariant features, ensuring close alignment among domains. Aligning domains at the feature level has been shown to significantly improve the efficacy of domain adaptation tasks. Local features capture localized information, such as edges and textures, which remain invariant across domains. As network depth increases, global features, which represent the entire image and highlight spatial correlations, also remain invariant across domains and are critical for distinguishing categories. Therefore, considering both local and global features when aligning domains at the feature level is necessary for effective domain adaptation. However, most current methods often focus on low layers to implement adaptation. For example, MSOSDA constructs each mini-batch to achieve a class-wise alignment between domains at the input layer [10]. These methods consider local features while neglecting global higher-level features and their rich semantic information correlations [11]. This leads to limitations in distinguishing some specific categories.

Therefore, the global feature extraction capability of the backbone network is crucial for the above adaptation task. Most RS scene classification methods use CNNs as backbones [12]. However, these have limitations due to the fixed receptive fields of local convolutional kernels, which restrict the modeling of medium- to long-range dependencies and overlook the global relational information within images. Many studies have addressed this by increasing the receptive field to capture global information through deeper CNN architectures, which introduces a substantial number of parameters. Recently, Transformers have been applied to image classification with promising results. In addition to pure Transformers, hybrid CNN–Transformer methods have been explored to leverage the advantages of both CNNs and Transformers [5]. However, effectively combining the two remains a critical challenge.

To address the above issues, we propose a CNN–Transformer with a consistency learning module and dynamic weighting strategy (Hy-MSDA), which can not only learn the invariant consistency features across domains by implementing an effective consistency learning module but also comprehensively consider the varying contributions of available source domains from both the perspectives of data distribution and model learning capability. To summarize, the primary contributions are as follows.

1) Toward the MUDA, the Hy-MSDA network is proposed, where the consistency learning module and dynamic weighting strategy operate at the feature level and decision level, respectively. They are closely interconnected and mutually reinforce each other to achieve collaborative cross-domain adaptation.
2) To further learn the invariant features across domains, a consistency learning module that roles in a hybrid CNN–Transformer is employed to enable the learning of invariant consistency features across domains, preserving the distribution consistency and retaining rich global semantic information.
3) To fully consider the varying contributions of available source domains, a dynamic weighting strategy is proposed from both the data distribution and model learning perspectives.
4) Experiments are conducted on several RS datasets involving classification and segmentation, and the experimental results demonstrate that Hy-MSDA is superior to other state-of-the-art methods.

II. Related Work

A. RS Classification Tasks

Considerable progress has been made in the field of RS image scene classification because of recent rapid advancements in deep learning techniques. Traditional approaches that rely on manual feature extraction have been largely replaced by deep learning, which automates the extraction of high-level semantic features from images. Classic CNN architectures, such as AlexNet [12], VGG [13], and ResNet [14], are fundamental deep learning models that have been widely adopted for RS image scene classification. Furthermore, researchers have made continual advancements in RS classification tasks via deep learning models. For example, attention-based network architectures have been proposed to improve image detail capture and classification accuracy in this task [15], 16]. Additionally, methods that incorporate multiscale and multimodal information have been developed to increase the robustness and accuracy of classification models [17], [18].

However, the performance achieved by these methods heavily depends on a large amount of accurately labeled data. In addition, the transferability of the models is not satisfactory; for example, the performance of the models on images obtained from different sensors tends to decrease [19]. Hence, challenges related to data annotation difficulties and limited model transferability still need to be addressed.

B. UDA Methods

UDA has gained significant attention in RS image classification tasks. By applying domain alignment techniques to labeled and unlabeled data, UDA methods enhance scene classification performance in the target domain [20], [21], [22]. Specifically, one way to implement UDA is to employ autoencoders to learn shared feature representations between both domains [23]. This is achieved by minimizing the difference in feature representations, reducing the distribution discrepancy, and improving the classification accuracy. However, this approach often encounters local optima, limiting the ability of the autoencoder to learn optimal feature representations. Consequently, the model may capture only local structures of input data and fail to learn higher-level semantic information and abstract features [24]. Another way to achieve UDA is to use generative adversarial networks (GANs) to generate synthetic target-domain samples [25]. By simultaneously making the generator network and the discriminator network learn the feature, GAN-based methods can generate samples whose distribution is similar to that of the target domain, enabling domain adaptation. However, these feature representations

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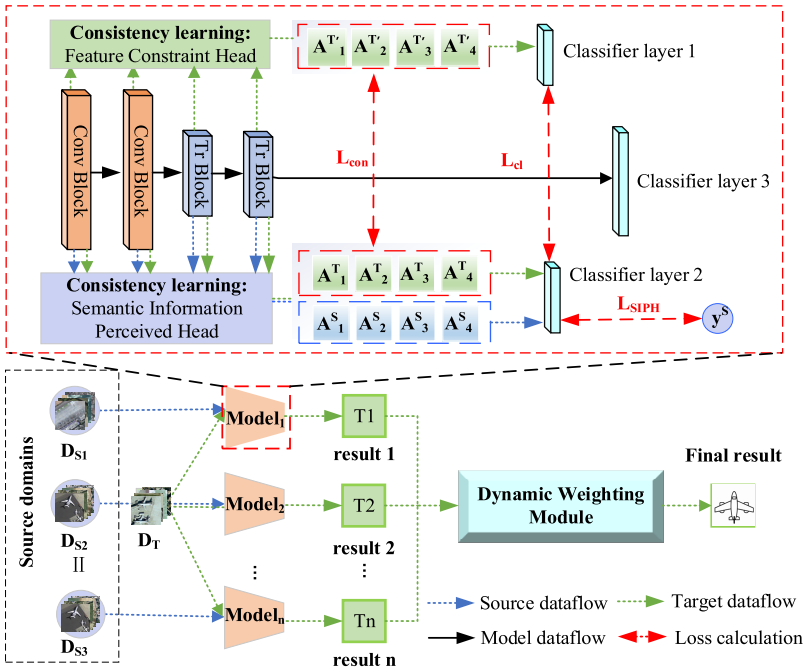


Fig. 1. Overview of our proposed Hy-MSDA.

generated by the generator may be less interpretable than those generated by nongenerative approaches.

Most UDA methods focus on aligning data distributions or local feature representations while neglecting the alignment of high-level semantic features [23]. This overlook can lead to the misalignment of certain categories in the target domain, resulting in classification errors that adversely impact remote sensing image classification performance.

C. MUDA Methods

Recently, there has been growing interest in MUDA methods, which leverage multiple source domains with labeled data to improve performance in the target domain with unlabeled data [5], [26]. In general, these methods can be classified into two primary types. One type involves merging all labeled domains into one domain and then utilizing UDA methods for model training. However, this mixing-based UDA approach may exacerbate model complexity when significant disparities exist among source domains [10], [27]. The other primary type of MUDA generally includes N models corresponding to N source domains, with different weights assigned to each source model. For example, MCC-DA is obtained by aggregating all models via parameter averaging in the model aggregation stage [28]. Through moment matching, M^{3}SDA aligns distributions across multiple source domains [29], whereas MDAN learns domain-invariant representations by employing multiple domain classifiers [27]. T-SVDNet emphasizes the importance of addressing differences between various source domains as well as the target domain [30]. Additionally, sample selection methods such as CMSS [31] select source-domain data that are more transferable and align them with target-domain data. In LtC-MSDA, graph neural networks are used to learn domain-invariant and discriminative representations [32]. TAMAN performs shared feature extraction, but it weights time and spatial features from different domains to assign videos [33].

However, it is worth noting that most existing methods do not fully exploit the complementary information across the rich multisource domains, considering the degree of contribution that every source domain makes to the unlabeled target-domain classification task.

III. Method

A. Overview

We denote n source domains and target domains as $D_S = \{(x_i^S, y_i^S)_{i=1}^{B_n}\}_{n=1}^{N}$ and $D_t = (x_i^T)_{i=1}^{B_t}$, where x_i^S and y_i^S represent the ith image and corresponding classification label of the nth source domain, respectively B_n represents the data volume of the nth source domain. The ith image in the target domain is represented as x_i^T, and the target-domain volume is B_t. Our objective is to train a model $F_\theta : x \to y$ using the labeled source domains D_S, which can effectively adapt to an unlabeled target domain D_t. Using multiple labeled sources D_S as inputs, Hy-MSDA trains a model that performs effectively on the unlabeled target domain D_t. As illustrated in Fig. 1, each source domain D_S has a corresponding network model M_n. The model M_n consists of several key components: a hybrid CNN–Transformer as a feature extractor, a consistency learning module, and three classifier layers, where all classifier layers are implemented through simple fully connected layers. The final model on the target domain is obtained by dynamically weighting the n models trained on the source domain according to our proposed dynamic weighting strategy.

B. Hybrid CNN–Transformer

To gain the high-level features containing global information, the proposed Hy-MSDA uses a CNN–Transformer as

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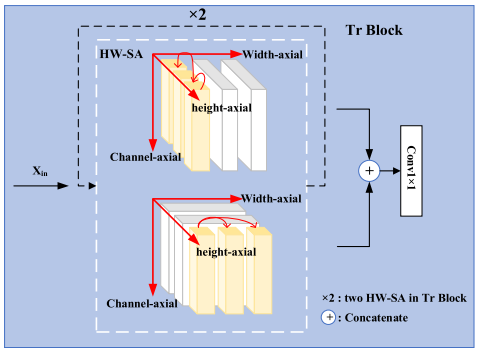


Fig. 2. Architecture of the Tr block. This block decomposes two HW-SAs, which contain self-attention at the height and width axes.

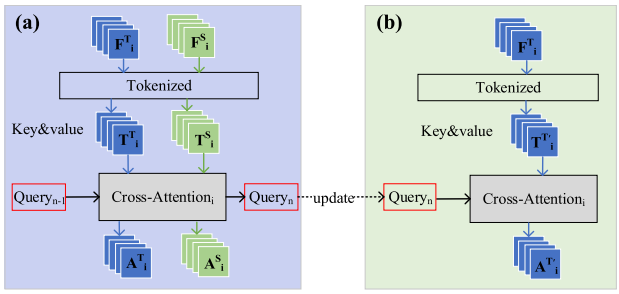


Fig. 3. Architecture of (a) semantic information perception head and (b) feature constraint head.

the hybrid backbone structure. As illustrated in Fig. 1, the CNN–Transformer network backbone combines the strengths of convolution layers in capturing localized details and the advantages of the Transformer in capturing distant relationships. First, the two blocks employ basic residual convolution blocks (Conv Block) that focus on modeling local details. Each Conv Block includes a downsampling layer to gradually reduce the feature resolution. The following two Transformer blocks (Tr Block) are subsequently employed to extract global features, capturing more complex and high-level semantic information.

However, the self-attention mechanism in Transformers introduces a significant increase in computational complexity because it requires computing pairwise relationships across all positions in the input sequence. This quadratic growth in computational complexity highlights the necessity of efficiently managing computational resources while maintaining model effectiveness and addressing this issue is crucial for CNN–Transformer models in classification tasks. Given the quadratic complexity of computations linked to self-attention mechanisms [34], we introduce HW-SA to efficiently capture global feature representations with comprehensive graph dependencies through lightweight computation, as depicted in Fig. 2. HW-SA applies independent axial self-attention operations in the height and width directions on the input features $X_{\text{in}} \in R^{h\times w\times c}$. The height, width, and channel of the feature are defined as h, w, and c, respectively. To perform height-axis self-attention, we project X_{in} into queries, keys, and values, denoted as Q, K, and V, respectively. The matrix of self-attention is then computed using l' as the scaling factor. To obtain the output, the normalized matrix softmax$(\cdot)$ is multiplied by V. The process of height-axial attention g_h is defined as follows:

$$g_h(X_{\text{in}}) = \text{softmax}\left(\frac{QK^T}{\sqrt{l'}}\right)V \tag{1}$$

where softmax$(\cdot)$ denotes the SoftMax operation.

The process of width–axial attention g_w, similar to g_h, captures information along the width axis of the input feature map. Then, we obtain the output X_1 of HW-SA$_1$

$$X_1 = g_h(X_{\text{in}}) + g_w(X_{\text{in}}). \tag{2}$$

Since performing a single HW-SA yields only pixel information in two directions, it fails to capture information from all the pixels and results in incomplete global information. Therefore, at least two serially connected HW-SA structures in Conv Blocks, referred to as HW-SA$_1$ and HW-SA$_2$, are used to obtain comprehensive contextual information and global dependencies from all the pixels in the entire image. The structures and computations of HW-SA$_1$ and HW-SA$_2$ are identical, as depicted in Fig. 2. The computational process is as follows:

$$X_2 = g_h(X_1) + g_w(X_1) \tag{3}$$

where X_2 is the output of HW-SA$_2$.

Then, before the output X_2 of HW-SA$_2$ is input into the subsequent architecture of the network, it undergoes a 1×1 convolution layer

$$X_{\text{out}} = \text{Conv}(X_2) \tag{4}$$

where X_{out} is the output of one Tr Block and Conv$(\cdot)$ is the convolution operation with a kernel size of 1.

Furthermore, to obtain richer and denser contextual information for RS scene classification tasks, the Hy-MSDA is equipped with two Tr Blocks as shown in Fig. 1. The decision to use two Tr Blocks was determined through hyperparameter experiments conducted in Section IV-B1, aimed at achieving the best performance. Notably, all HW-SAs within a source-domain model share parameters to minimize the number of additional parameters.

C. Consistency Learning

To enhance the learning of domain-invariant features and identify semantically related areas at various representation levels, we propose a consistency learning module consisting of two components: the semantic information perceptual head (SIPH) and the feature constraint head (FCH), as illustrated in Fig. 3.

Specifically, for perceiving global high-level features and rich semantic information better, we define a learnable SIPH as a query, aimed at capturing the statistical information of categories. Subsequently, we utilize SIPH to compute attention maps over the features extracted from the backbone. The final prediction is obtained by applying a small convolution to the attention maps. The trained SIPH then guides the learning of the FCH for domain adaptation.

As illustrated in Fig. 3, we represent the intermediate features $\{F_n^S, F_n^T\}$ at different scales $n \in (1, 2, 3, 4)$ as tokens

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and feed them as input Keys and Values to a series of cross-attention blocks [35] in the SIPH. At the first scale, the initialized perceptual head Query_0 interacts with the token features $\{T_n^S, T_n^T\} \in R^{4C\times(H/16\times W/16)}$, obtaining the perceptual head Query_1 and semantic perceptual attention map $\{A_1^S, A_1^T\}$. After adjusting the token dimension of the next layer with the 1×1 convolution, the next cross-attention generates the semantic perception attention map at scale 2, and so on. These attention maps $\{A_n^S, A_n^T\}$ can be used with a classification layer to obtain the final prediction results $\{P_n^S, P_n^T\}$ at each scale. For simplicity, we illustrate the cross-attention updating process using the tokenized features $\{T_n^S\}$ at scale 1 and the initialized Query_0 as follows:

$$q = QW_Q, \quad k = T_1^S W_K, \ v = T_1^S W_V \tag{5}$$

$$\text{CA}\left(Q, T_1^S\right) = \text{softmax}\left(\frac{qk^T}{\sqrt{l}}\right)v \tag{6}$$

where CA and Q denote cross-attention and Query, respectively, W_K, W_V, and W_Q are the parameter matrices for linear projection. The multihead cross-attention (MCA) is the extension with N independent cross-attentions and projects their concatenated outputs as follows:

$$\text{MCA}\left(Q, T_1^S\right) = C\left(\text{CA}_i\left(Q, T_1^S\right), i \in (1, \ldots, N)\right) \tag{7}$$

where C denotes the concatenation operation. Here, we can obtain the attention maps A_1^S extracted from multiheads. Finally, the Q can be updated by

$$Q_1 = \text{MCA}\left(Q, T_1^S\right) + Q \tag{8}$$

$$\text{Query}_1 = \text{Conv}(\text{MLP}(\text{Norm}(Q_1)) + Q_1) \tag{9}$$

where Query_1 indicates the updated proxy at scale 1. Conv is a 1×1 convolutional layer to reduce the channel dimension. It is worth noting that T_1^T does not affect the updating of the learnable parameter Query during backpropagation, while T_1^S does. Moreover, the other scales are also the same as the above equations.

To ensure strong discrimination between different semantic categories, we employ cross-entropy loss L_{ce} to supervise the prediction results P_n^S of the source domain with the corresponding label y_i as follows:

$$L_{\text{SIPH}} = \frac{1}{n}\sum_n L_{\text{ce}}\left(\text{softmax}\left(P_n^S\right), y_i\right) \tag{10}$$

where $n \in (1, 2, 3, 4)$ represents different scales and where y_i is the classification label of the source domain.

The MMD loss is employed to update model parameters, ensuring alignment of feature distributions between the source and target domains. The goal of MMD is to minimize the expected difference in the feature space between the source and target domains, thereby making the feature representations of the two distributions more similar. Therefore, Hy-MSDA effectively mitigates domain shift effects and enhances the model's generalization capability on the target domain. The formula is as follows:

$$L_{\text{ST}} = \text{MMD}\left(A_n^S, A_n^T\right) = \left\|\mu_{A_n^S} - \mu_{A_n^T}\right\|_{\text{H}} \tag{11}$$

where $||\cdot||_{\text{H}}$ denotes the norm in the feature space.

The FCH receives Query_n from the SIPH, which contains rich semantic information of different categories. Using these methods, the FCH generates semantic-guided attention maps $A_n^{T'}$ that reveal semantically related regions at various representation levels. Considering the perturbation during this process, the learned features from the SIPH and FCH methods may exhibit inconsistency when provided with the same unlabeled target-domain input samples. We leverage the aforementioned inconsistency to constrain discriminative features, which is important for learning domain-invariant features. Specifically, we devise a consistency regularization between A_n^T from the SIPH and $A_n^{T'}$ from the FCH at different scales. This regularization enhances the robustness of global semantic representations by leveraging unlabeled target-domain data. We formulate the loss via the mean squared error (MSE) as follows:

$$L_{\text{TT}'1} = \frac{1}{n}\sum_n L_{\text{MSE}}\left(\text{softmax}\left(A_n^T\right), \text{softmax}\left(A_n^{T'}\right)\right) \tag{12}$$

where $L_{\text{TT}'1}$ represents the consistency loss between the classification results of the target domain from SIPH and FCH.

In addition, similar to SIPH, the generated attention maps are used to guide the classification result C_n^T within FCH. However, this process only involves data from the unlabeled target domain. By applying cross-entropy loss to the predictions C_n^T and P_n^T, we can effectively improve the model's domain adaptation performance by leveraging both the data and the unlabeled data from the source and target domains

$$L_{\text{TT}'2} = \frac{1}{n}\sum_n L_{\text{ce}}\left(\text{softmax}\left(C_n^T\right), \text{softmax}\left(P_n^T\right)\right) \tag{13}$$

where $L_{\text{TT}'2}$ represents the consistency loss between the classification results of the target domain from SIPH and FCH.

So, the consistency loss between the target and generated target attention maps is calculated as follows:

$$L_{\text{TT}'} = L_{\text{TT}'1} + L_{\text{TT}'2}. \tag{14}$$

D. Dynamic Weighting Strategy

Considering the variations in contributions from different source domains, we propose a dynamic weighting strategy that takes two factors into account. The first factor is the similarity between the sample distributions of each source domain and the target domain, as shown in Fig. 4(a). The similarity quantifies how closely the samples from a source domain resemble those from the target domain. Specifically, the greater the similarity is, the more significant the contribution of that source to the task. The second factor is the models' ability to learn invariant-domain features, where the models are trained with different source domains and the target domain, as depicted in Fig. 4(b). We assess Hy-MSDA and assume that the models with stronger learning capabilities are assigned higher weights.

By combining these two factors, we can assign dynamic weights to the source-domain models in domain adaptation tasks, which allows us to fully utilize the rich information contained within multiple source domains, enhancing the overall domain adaptation performance.

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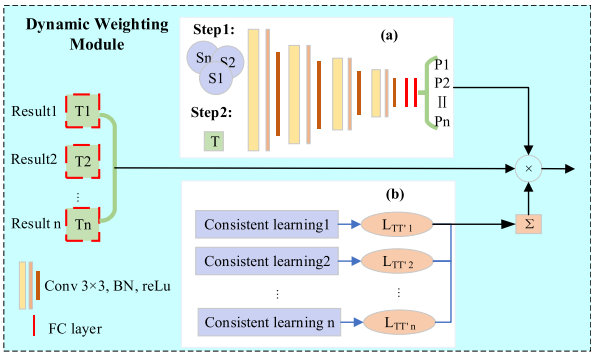


Fig. 4. Architecture of the dynamic weighting strategy. It contains two factors: (a) similarity of sample distributions and (b) learning capabilities of multiple source models.

1) Dynamic Weighting Based on the Similarity of Sample Distributions: To explore the contributions of different source domains to the domain adaptation task of the target domain, we initially focused on the similarity of sample distributions across different domains. We introduce an N-class classifier that has been pretrained using n source domains along with their corresponding domain labels, as illustrated in Fig. 4(a) step 1. During the training of Hy-MSDA, in step 2, the target-domain samples are input into the classifier, yielding probability values indicating the likelihood of the target domain being classified as each of the N source domains. These probability values $\{P_1, P_2, \ldots, P_n\}$ reflect the similarity between the sample distributions of N source domains and one target domain. Then, we weigh the results obtained from the N models on the basis of these probabilities to obtain the final classification result for the target samples. Mathematically, this process is defined as follows:

$$\{\text{Out}_1, \text{Out}_2, \ldots, \text{Out}_n\} = \{T_1 \cdot P_1, T_2 \cdot P_2, \ldots, T_n \cdot P_n\} \quad (15)$$

where $\{\text{Out}_1, \text{Out}_2, \ldots, \text{Out}_3\}$ represents the classification results of the target domain after applying the first weighting, $\{T_1, T_2, \ldots, T_n\}$ represents the outputs from N source models, and $\cdot$ denotes the multiplication operation.

2) Dynamic Weighting Based on the Learning Capabilities of Multiple Source Models: Given that each model M_n operates independently and has unique parameters and varying learning capabilities, we consider each model's ability to learn domain-consistent features. To enhance the performance through a more effective weighting strategy, we tightly integrate the weighting process with the consistency learning module. Specifically, we employ $L_{\text{TT}'}$ introduced in Section III-C as an indicator of each model's learning capability. Here, $L_{\text{TT}'}$ represents the consistency loss between the target and generated target-domain features at each scale. As illustrated in Fig. 4(b), we derive N consistency losses on the basis of N source domain models, denoted as $\{L_{\text{TT}'_1}, L_{\text{TT}'_2}, \ldots, L_{\text{TT}'_n}\}$, where $n \leq N$. Therefore, after the weighting strategy that is based on the similarity of the sample distribution, we calculate the final classification results. The calculation process is as follows:

$$\text{Output}_T = \frac{1/L_{\text{TT}'_1}}{\sum_{n=1}^{N} 1/L_{\text{TT}'_n}} \cdot \text{Out}_1 + \cdots + \frac{1/L_{\text{TT}'_n}}{\sum_{n=1}^{N} 1/L_{\text{TT}'_n}} \cdot \text{Out}_n \quad (16)$$

Algorithm 1 Hy-MSDA

Input: The initialized networks, a pretrained N-class classifier, N labeled source domains datasets D_s and one unlabeled target dataset D_t, and hyperparameters: $batchsize, lr, \alpha 1, \alpha 2, \alpha 3$.
Output: A trained network and prediction results.
1: Step1: Training
2: for n = 1 to N **do**
3: **for** i = 1 to K **do**
4: Divide the train datasets D_s and D_t into K batches.
5: train the network M_n via Equation. (17).
6: **then**
7: Compute the target result from N networks $M_1, \ldots, M_n$ via Equation. (15) and (16).
8: end
9: Step2: Testing
10: Evaluate the models via Equation. (18)–(21) and store the target results.

where Output_T represents the classification results of the target domain after the second weighting. A smaller $L_{\text{TT}'}$ indicates a stronger ability of the model to learn domain-invariant features. To reflect this, we utilize the reciprocal of $L_{\text{TT}'}$ in (16).

E. Loss Function

By combining the aforementioned losses, the proposed Hy-MSDA can be formulated as follows:

$$\min_{(M_1, M_2, \ldots, M_n)} L_{\text{SIPH}} + \alpha_2 L_{\text{TT}'} + \alpha_3 L_{\text{ST}} \quad (17)$$

where α_2 and α_3 are used for balancing the contributions of the different losses. Additionally, Hy-MSDA is trained and evaluated as Algorithm 1.

IV. Experiment

A. Dataset and Experimental Design

1) Experimental Benchmark Datasets: Four remote sensing datasets for scene classification, namely, AID [36], UC Merced Land-Use [37], PatternNet [38], and NWPU [39], are used to verify the proposed method. We refer to these datasets as s1, s2, s3, and s4, respectively. The four datasets contain 30, 21, 45, and 28 categories, respectively. Since there are some shared categories among the four datasets, we selected ten common classes and standardized their names. Fig. 5 presents some representative samples from each dataset.

In addition, three RS datasets for image segmentation, namely, the aerial image segmentation dataset [40], the Zurich summer dataset [40], and the WHDLD dataset [41] are utilized to implement the land-cover segmentation experiment. These additional experiments aimed to prove the effectiveness of Hy-MSDA on the segmentation task. To simplify the introduction of these experiments, we refer to the aerial image segmentation dataset, Zurich summer, and WHDLD as n1, n2, and n3, respectively. The specific information about these experimental datasets is shown in Table I. Moreover,

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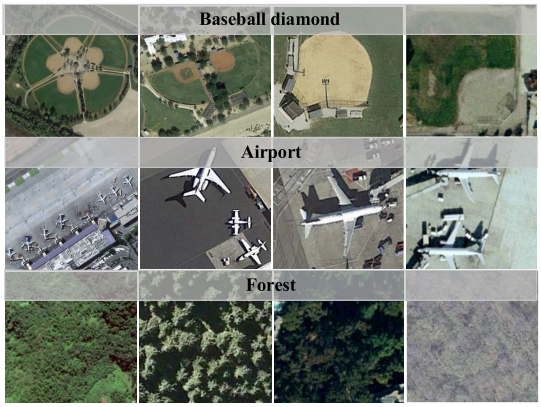


Fig. 5. Representative samples in four remote sensing datasets for scene classification.

TABLE I
DETAILED INFORMATION ABOUT THE EXPERIMENTAL DATASETS FOR CLASSIFICATION AND SEGMENTATION

Tasks	Datasets	Spatial Resolution	Classes	Dataset Scale
Scene classification	AID (s1)	0.5 m-8 m	10	8k
	Merced (s2)	0.3 m		
	PatternNet (s3)	0.062 m–4.693 m		
	NWPU (s4)	0.2 m–30 m		
Land-Cover segmentation	Aerial Image Segmentation Dataset (n1)	0.5 m	3	8k
	Zurich Summer (n2)	0.62 m		
	WHDLD (n3)	2 m		
Cloud detection	Landsat-8 (d1)	30 m	2	8k
	Sentinel-2 (d2)	10 m		
	GF-2 (d3)	4 m		

we also conduct experiments on three cloud datasets, namely Landsat-8 [42] [43], Sentinel-2 [44], and GF-2 [45]. We refer to them as d1, d2, and d3, respectively.

2) Evaluation Metrics: To evaluate the performance of Hy-MSDA, we adopted several widely used evaluation metrics, which are calculated via the following equations:

$$\text{Accuracy} = \frac{1}{N}\sum_{i=1}^{N}\frac{\text{TP}_i + \text{TN}_i}{\text{TP}_i + \text{TN}_i + \text{FP}_i + \text{FN}_i} \tag{18}$$

$$\text{Precision} = \frac{1}{N}\sum_{i=1}^{N}\frac{\text{TP}_i}{\text{TP}_i + \text{FP}_i} \tag{19}$$

$$\text{Recall} = \frac{1}{N}\sum_{i=1}^{N}\frac{\text{TP}_i}{\text{TP}_i + \text{FN}_i} \tag{20}$$

$$F1 = \frac{2 \times \text{Recall} \times \text{Precision}}{\text{Recall} + \text{Precision}} \tag{21}$$

where N is the number of classes, TP_i is the number of true positives for the i class, TN_i is the number of true negatives for the i class, FP_i is the number of false positives for the i class, and FN_i is the number of false negatives for i class.

TABLE II
METRICS OF THE ABLATION STUDY FOR THE PROPOSED Hy-MSDA ON THE RS SCENE CLASSIFICATION DATASET (s1, s2, s3 → s4). ↑ INDICATES THAT THE LARGER THE METRIC VALUE IS, THE BETTER THE PERFORMANCE. THE **Bold** TEXT INDICATES THE BEST RESULTS

Network	CNN	√				
	C-T		√	√	√	√
Consistency learning				√	√	√
Dynamic Weighting strategy	a*				√	√
	b*					√
Accuracy↑		0.873	0.907	0.939	0.954	**0.977**
Precision↑		0.881	0.920	0.945	0.959	**0.982**
Recall↑		0.875	0.926	0.941	0.955	**0.973**
F1 Score↑		0.882	0.914	0.933	0.948	**0.966**

In addition to these classification metrics, we also employed the mean intersection over union (mIoU) metric to evaluate the performance of our segmentation tasks as follows:

$$\text{mIoU} = \frac{1}{k}\sum_{i=1}^{k}\frac{\text{TP}_i}{\text{TP}_i + \text{FP}_i + \text{FN}_i} \tag{22}$$

where TP_i, FP_i, and FN_i are defined similarly to (18) and where k is the number of classes.

3) Implementation Details: All the experiments were conducted on an Ubuntu system equipped with NVIDIA RTX4090 GPUs and implemented via PyTorch. The training settings used in our experiments were as follows: the number of epochs was set to 100, the optimizer used was SGD with a momentum of 0.9, the learning rate was set to $2 \times e^{-4}$, and the weight decay was $1 \times e^{-8}$. The batch size was determined to be 8 on the basis of the available amount of data. Additionally, the number of samples was set to 800 per class for scene classification datasets.

B. Ablation Experiments of Hy-MSDA

To evaluate the impact of each major component of Hy-MSDA, we conducted several ablation experiments. In Hy-MSDA, three major components are employed to improve performance: CNN–Transformer, consistency learning, and a dynamic weighting strategy. By removing each of them from the model, we performed several ablation studies to validate their effectiveness. We utilized s1, s2, and s3 as the source domains and s4 as the target. The results are shown in Table II, where "√" indicates the inclusion of the component. "CNN" and "C-T" represent the CNN backbone and hybrid CNN–Transformer backbone, respectively. Additionally, "a*" and "b*" represent two weighting strategies based on the similarity of sample distributions and the learning capabilities of multiple source models, respectively.

1) Effectiveness of the Hybrid CNN–Transformer Network: First, we conducted experiments to assess and compare the efficacy of the hybrid CNN–Transformer network against a pure CNN model, which served as our baseline. The results demonstrate that the CNN–Transformer is superior to the CNN baseline, as shown in Table II, highlighting its effectiveness

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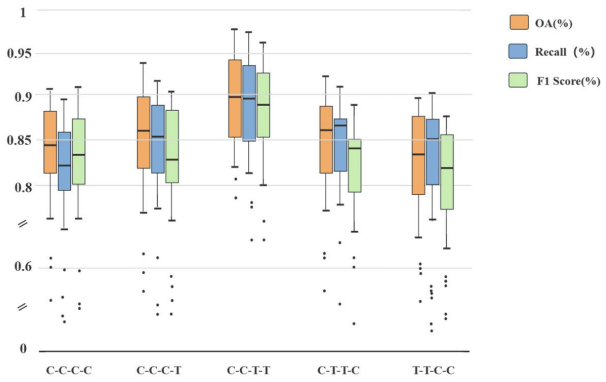


Fig. 6. Effectiveness of the position and number of Tr blocks in the CNN–Transformer backbone.

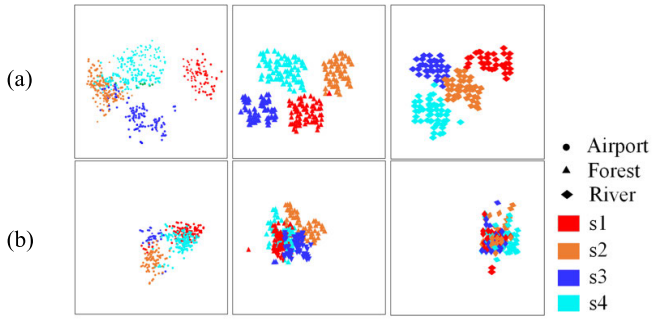


Fig. 7. t-SNE visualization of features (a) before and (b) after being processed by the consistency learning module. The points are represented by different colors and shapes to indicate their corresponding domains and scene categories, respectively.

for RS image classification tasks. Specifically, the hierarchical feature extraction capabilities of CNNs are enhanced by the self-attention of transformers, which makes the model better able to capture long-range dependencies and implicit global semantic relations within the image. This combination of global and local information leads to more accurate predictions and a significant increase in classification performance.

Furthermore, to further explore the effectiveness of the hybrid CNN–Transformer, we researched the architecture composition. The network consists of four blocks, with a detailed exploration of the position and quantity of Conv Blocks and Tr Blocks. We performed five experiments to investigate the optimal configuration of the CNN–Transformer backbone. Specifically, "C-C-C-T" denotes a network with three Conv Blocks and one Tr Block, positioned last. Similar interpretations apply to configurations for "C-C-T-T," "C-T-T-C," and "T-T-C-C." To evaluate the performance of these configurations, we used a box plot to visualize three evaluation metrics: OA, Recall, and the $F1$ score. Fig. 6 presents the distributions of the evaluation metrics for the performances of the five configurations. The results depicted in Fig. 6 show that the "C-C-T-T" configuration achieves higher metrics and results in fewer outliers in terms of accuracy. This suggests that the configuration with two Conv Blocks followed by two Tr Blocks positioned at the end of the network yields the best performance in terms of classification accuracy.

2) Effectiveness of Consistency Learning: Then, we conducted experiments to prove the effectiveness of the consistency learning module in Hy-MSDA. The experiments were performed via our method with and without the consistency learning module. The results in Table II indicate a noteworthy improvement in the segmentation performance across various categories when the consistency learning module is incorporated into the CNN–Transformer networks. Specifically, the consistency learning module improves the accuracy by 3%.

To further illustrate the ability of the consistency learning module to align features from different domains, we conducted the MUDA experiment from s1, s2, s3 to s4, where we randomly selected three categories out of ten and visualized them via t-SNE, as shown in Fig. 7(a) and (b), which represent the distributions before and after the consistency learning module, respectively. After consistency learning, regardless of the category, the originally distinct feature distributions from different domains become indistinguishable. This suggests that the consistency learning module effectively filters out domain-specific features and aligns domain-invariant features, which enables the model to generalize well across different domains and improve its performance in segmenting images from the target domain.

3) Effectiveness of the Dynamic Weighting Strategy: The proposed dynamic weighting strategy includes two measurements: sample distribution and model learning ability. To verify the effectiveness of the dynamic weighting strategy, we assessed the performance of the dynamic weighting strategy through two ablation experiments. The results in Table II demonstrate that the inclusion of the dynamic weighting strategy based on sample distribution improved the performance of Hy-MSDA, with the accuracy increasing from 0.939 to 0.954. Furthermore, the addition of the dynamic weighting strategy based on the model's learning ability further improved the accuracy of Hy-MSDA by up to 2.3%.

The relatively superior performance of the dynamic weighting strategy can be attributed to its consideration of both the data distribution and the learning ability of source models, which allows for a more comprehensive way to handle multiple source domains and their corresponding models. Specifically, by considering the data distribution, the dynamic weighting strategy assigns a larger weight to the source model whose data distribution is closer to that of the target domain. Additionally, the dynamic weighting strategy incorporates the model's learning ability, which weights multiple outcomes on the basis of the different abilities of each source domain model to learn consistency features. Considering both the data distribution and the model learning ability, the Hy-MSDA method achieves better classification performance.

C. Comparison With Mainstream State-of-the-Art Methods

To assess the effectiveness of Hy-MSDA, we implemented several experiments on RS scene classification datasets and compared the performance of Hy-MSDA with that of six state-of-the-art MUDA methods, namely, M^3SDA [29], MFSAN [9], LCt-MSDA [32], T-SVDNet [30], RRL [46], PTMDA [47], MCC-DA [28], and SUMDA [48].

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TABLE III

ACCURACIES OF HY-MSDA AND SIX STATE-OF-THE-ART METHODS ON THE RS SCENE CLASSIFICATION DATASETS. **Bold Red** INDICATES THE BEST, AND **Bold Blue** INDICATES THE SECOND BEST. THE LARGER THE VALUE IS, THE BETTER THE PERFORMANCE OF THE METRIC

Domain \ Method	M^3SDA	MFSAN	LCt-MSDA	T-SVDNet	MCC-DA	PTMDA	SUMDA	RRL	Hy-MSDA
(s1→s2)	0.887	0.912	0.873	0.854	0.940	0.944	0.944	**0.946**	**0.959**
(s2→s3)	0.879	0.910	0.870	0.859	0.938	0.930	0.933	**0.933**	**0.947**
(s3→s4)	0.870	0.907	0.868	0.855	0.931	0.928	**0.939**	0.937	**0.953**
(s1, s3→s2)	0.883	0.919	0.890	0.865	0.950	**0.951**	0.949	0.949	**0.968**
(s1, s4→s2)	0.895	0.920	0.905	0.860	0.945	0.948	0.950	**0.951**	**0.972**
(s1, s2→s3)	0.914	0.938	0.905	0.881	0.966	0.965	**0.968**	0.964	**0.979**
(s1, s4→s3)	0.897	0.904	0.896	0.879	0.960	**0.963**	0.959	0.961	**0.968**
(s1, s3→s4)	0.920	0.931	0.901	0.876	0.960	0.957	0.960	**0.965**	**0.980**
(s2, s3→s4)	0.917	0.940	0.908	0.881	0.967	**0.968**	0.965	0.967	**0.978**
(s1, s3, s4→s2)	0.901	0.923	0.898	0.869	0.957	0.954	0.956	**0.959**	**0.976**
(s1, s2, s4→s3)	0.916	0.940	0.911	0.885	0.961	**0.967**	0.962	0.965	**0.980**
(s1, s2, s3→s4)	0.922	0.928	0.886	0.884	**0.964**	0.960	0.963	0.955	**0.977**

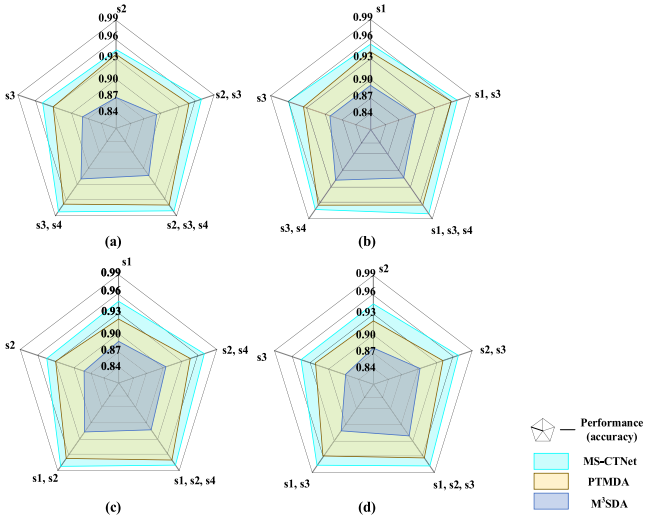


Fig. 8. Radar charts of the experimental results of Hy-MSDA, PTMDA, and M^3SDA in terms of the RS scene classification performance metric (accuracy) on different target domains: (a) s1, (b) s2, (c) s3, and (d) s4.

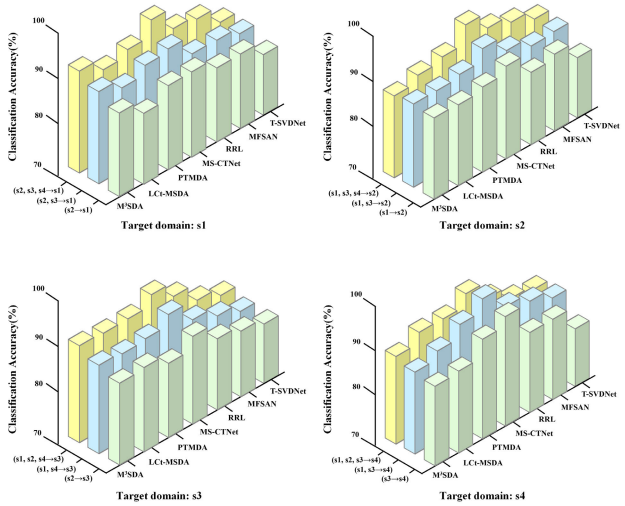


Fig. 9. Histograms of the accuracies of Hy-MSDA and six state-of-the-art methods on the RS scene classification datasets with different target domains: (a) s1, (b) s2, (c) s3, and (d) s4.

The experimental results are presented via the accuracy metric in Table III. We can see that Hy-MSDA significantly outperforms all the other methods in almost all the cases on the four datasets, improving the accuracy by up to 6%. This proves that by integrating a hybrid CNN–Transformer network with consistent learning and dynamic weighting, Hy-MSDA can achieve better RS scene classification performance. Specifically, our proposed consistency learning module effectively captures and assimilates valuable information from diverse source domains, thereby enhancing the network's ability to perceive and learn domain-invariant features. This allows Hy-MSDA to effectively generalize its learning to the target domain. Additionally, the dynamic weighting strategy assigns varying weights to the advantages of distinct source domains, prioritizing beneficial information while mitigating the impact of irrelevant noise. This dynamic weighting further improves the discriminative power of Hy-MSDA in RS scene classification.

Additionally, we visualized the results of our method compared with those of PTMDA and M^3SDA in Fig. 8. The radar charts of the experimental results of Hy-MSDA, PTMDA, and M3SDA in terms of the RS scene classification performance metric (accuracy) on different target domains: (a) s1, (b) s2, (c) s3, and (d) s4.

Fig. 9 shows histograms depicting the accuracy results of Hy-MSDA and six state-of-the-art methods on the RS scene classification datasets with different target domains: s1, s2, s3, and s4. Our method exhibits superior accuracy across one, two or three source domains, demonstrating its effectiveness in handling domain shifts. Furthermore, the use of three source domains yields the best results for all methods, highlighting the benefits of leveraging more source domains. The increased richness of information from multiple sources enables models to learn more robust and transferable features, resulting in enhanced classification accuracy across various target domains.

The four figures represent the domain adaptation experiments conducted on target domains s1, s2, s3, and s4. Within each figure, the results using one, two, and three source

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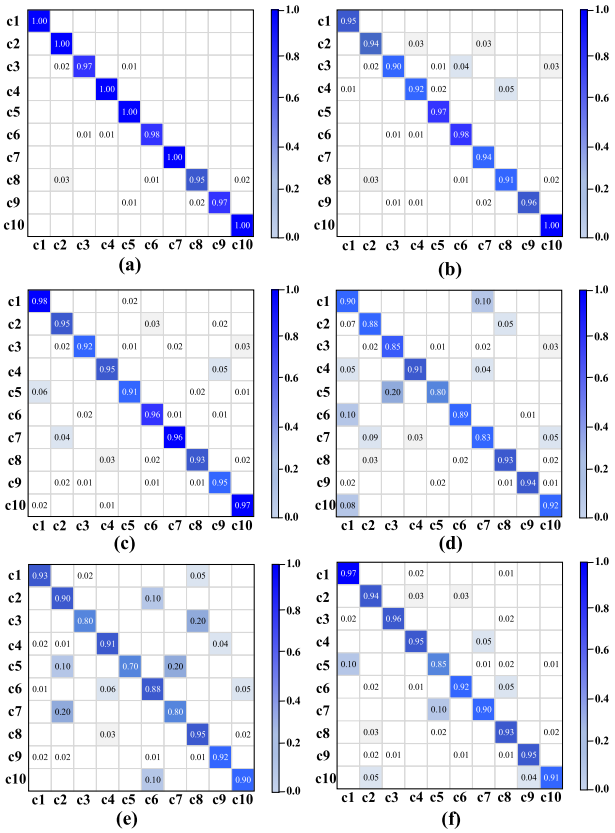


Fig. 10. Visualization of the confusion matrices of different methods on the scene classification task from s1, s2, s4 to s3. (a) Hy-MSDA, (b) RRL, (c) T-SVDNet, (d) LCt-MSDA, (e) M^{3}SDA, and (f) MFSAN.

domains are depicted. From the visualization, it is evident that the proposed method consistently outperforms the other methods in each experiment. The performance improvement is visible across all target domains, which utilize different numbers of source domains. This further validates the effectiveness of Hy-MSDA in addressing the domain adaptation problem in RS scene classification.

Fig. 9 shows histograms depicting the accuracy results of Hy-MSDA and six state-of-the-art methods on the RS scene classification datasets with different target domains: s1, s2, s3, and s4. Our method exhibits superior accuracy across one, two or three source domains, demonstrating its effectiveness in handling domain shifts. Furthermore, the use of three source domains yields the best results for all methods, highlighting the benefits of leveraging more source domains. The increased richness of information from multiple sources enables models to learn more robust and transferable features, resulting in enhanced classification accuracy across various target domains.

To further illustrate the effectiveness of the proposed Hy-MSDA, we conducted domain adaptation with s3 as the target domain, and then we visualized the confusion matrices of six different methods, where c1–c10 indicate the aforementioned ten shared categories among the four datasets. As we can see from Fig. 10, Hy-MSDA outperforms the other methods in each category. Specifically, Hy-MSDA significantly reduces classification errors, leading to a remarkable improvement in accuracy in the forest category, from 0.70 to 1.00. In addition, substantial improvements in accuracy were observed across other categories. In summary, all the experimental results indicate that Hy-MSDA achieves remarkable performance in scene classification tasks.

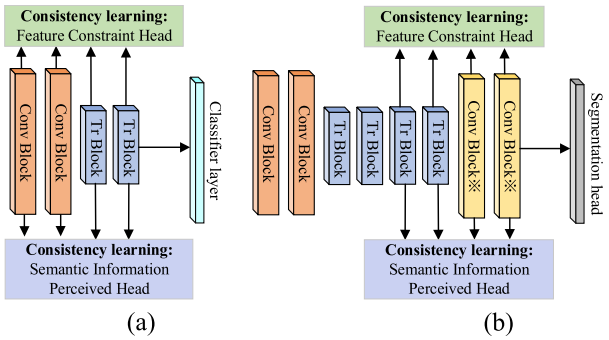


Fig. 11. CNN–Transformer backbone for (a) RS scene classification task and (b) RS segmentation task.

D. Validation of the Universality of Hy-MSDA in RS Image Segmentation Experiments

To investigate the universality of Hy-MSDA, we conducted experiments on RS land-cover segmentation datasets. To adapt Hy-MSDA for segmentation tasks, we made partial adjustments to the network architecture. Specifically, we retained the overall network design but changed it to a U-shaped framework to meet the requirements of the segmentation task. Fig. 11 shows the original network (a) for classification and the modified network (b) for segmentation. In segmentation network (b), we replaced the original classification layer with a segmentation head, which consists of a simple 1×1 convolution. Additionally, each Conv Block in the U-shaped structure includes an upsampling layer to gradually restore the resolution of the features.

To verify the generalizability of our method, we perform segmentation experiments on RS land-cover segmentation datasets using the Hy-MSDA model. Specifically, we conducted the proposed Hy-MSDA with n1, n2, and n3 as the target domain, containing the cases of one and two source domains. Moreover, we also conducted single-domain supervised segmentation experiments on n1, n2, and n3 via the Segformer [49] model. The experimental results are presented in Table IV.

As indicated in Table IV, when performing land-cover segmentation with one source domain and one target domain [i.e., (n2 $\rightarrow$ n1), (n3 $\rightarrow$ n1), (n1 $\rightarrow$ n2), (n3 $\rightarrow$ n2), (n1 $\rightarrow$ n3)], the experimental results demonstrate that Hy-MSDA is a promising method for land-cover segmentation, even without the use of target-domain label information. Specifically, we observed that Hy-MSDA underperformed the Segformer by up to 0.6% in terms of the mIoU, whereas the Segformer was trained with n1, n2, and n3. Compared with the Segformer, which is trained with target-domain labels, Hy-MSDA performs slightly worse in one source domain and one target domain. However, in the two source domains and one target domain, Hy-MSDA achieves results comparable to those of Segformer's supervised learning approach.

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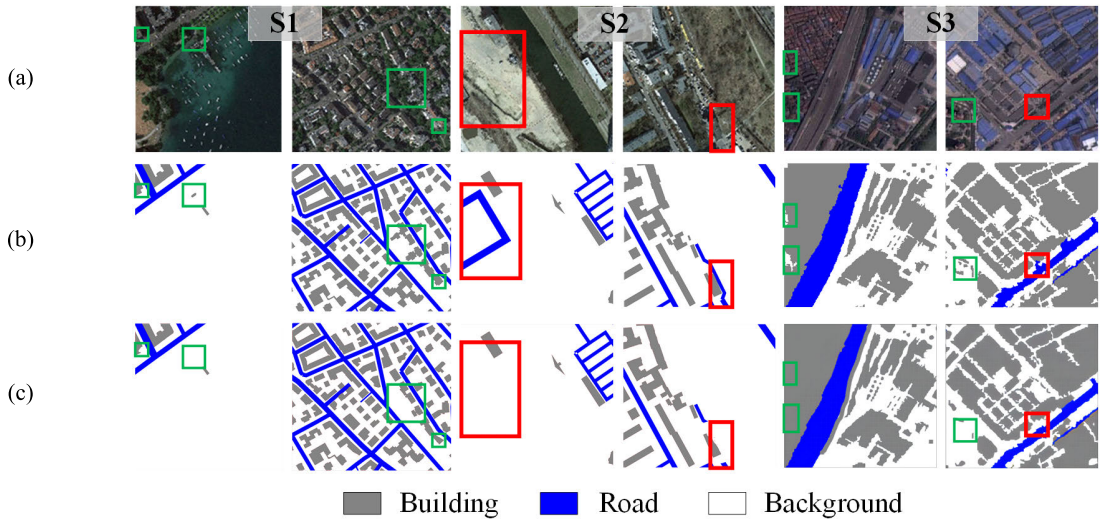


Fig. 12. Visualized segmentation results of Hy-MSDA on the land-cover segmentation datasets (n1, n2 → n3): (a) images, (b) corresponding labels, and (c) segmentation results. The red bounding boxes and green bounding boxes indicate buildings and roads where Hy-MSDA's segmentation results diverge significantly from the corresponding labels.

TABLE IV

ACCURACIES OF Hy-MSDA AND THE SEGFORMER ON THE RS LAND-COVER SEGMENTATION DATASETS. THE LARGER THE VALUE IS, THE BETTER THE PERFORMANCE OF THE METRIC

Method / Domain	Hy-MSDA			Segformer		
	m (%) [14]	B (%)	R (%)	m (%)	B (%)	R (%)
n2→n1	81.7	83.3	79.2			
n3→n1	82.1	83.4	80.3	85.2	86.1	83.7
n2, n3→n1	87.4	90.1	86.6			
n1→n2	82.0	83.9	80.7			
n3→n2	82.5	84.0	79.9	88.5	89.8	87.6
n1, n3→n2	88.2	90.4	87.0			
n1→n3	83.1	84.2	81.1			
n2→n3	82.9	83.9	81.7	91.2	93.1	88.5
n1, n2→n3	90.1	92.8	87.3			

Note: m indicates the mIou value, and B and R represent the accuracy values of Building and Road, respectively.

TABLE V

METRICS OF THE ABLATION STUDY FOR THE PROPOSED Hy-MSDA ON THE RS LAND-COVER SEGMENTATION DATASETS (N1, N2 → N3). ↑ INDICATES THAT THE LARGER THE METRIC VALUE IS, THE BETTER THE PERFORMANCE. THE **Bold** TEXT INDICATES THE BEST RESULTS

Network	CNN	√				
	C-T		√	√	√	√
Consistency learning				√	√	√
Dynamic Weighting strategy	a*				√	√
	b*					√
mIou↑		0.829	0.842	0.877	0.893	**0.901**

Furthermore, we conducted a series of ablation experiments on the land-cover segmentation dataset to comprehensively demonstrate the effectiveness of the proposed method across its various components. As illustrated in Table V, the experimental results reveal that Hy-MSDA, which integrates both consistent learning and dynamic weighting strategies, achieved the highest mIoU score compared to other configurations. This significant performance improvement not only proves the advantages of our approach but also validates the individual contributions of the proposed components in enhancing segmentation accuracy. By analyzing the impact of each element, we can better understand how consistent learning aids in maintaining domain-invariant features while dynamic weighting optimally balances information from diverse data sources. These findings highlight the robustness of Hy-MSDA in effectively addressing the challenges posed by land-cover segmentation tasks.

The visualized results in Fig. 12 further support the effectiveness of Hy-MSDA. Specifically, as shown in Fig. 12, Hy-MSDA yields segmentation results that are generally consistent with the corresponding labels, correctly classifying the majority of buildings and roads. However, slight deviations are observed in the vicinity of buildings with shadows and irregular shapes, as indicated by the green bounding boxes. Additionally, there are instances of erroneous road segmentation in certain scenes, as shown in the red boxes. Overall, Hy-MSDA has the potential as a universal segmentation method for land-cover classification, demonstrating its ability to utilize multiple source domains and achieve competitive segmentation results.

Additionally, we validated the effectiveness of Hy-MSDA on cloud detection tasks using the Landsat-8, Sentinel-2, and GF-2 datasets as target domains. The quantitative and qualitative results of our experiments are presented in Table VI and Fig. 13. We also employed Segformer as a comparative network to validate our cloud detection results. From the tables, it is evident that as the number of source domains increases, the accuracy of cloud detection improves. Moreover, our cloud detection results closely approximate the labeled

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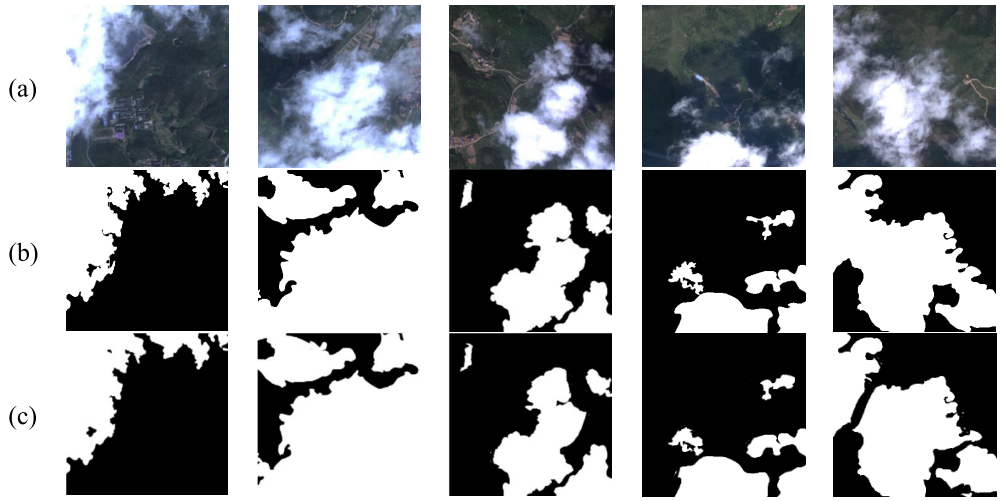


Fig. 13. Visualized segmentation results of Hy-MSDA on the cloud detection datasets (d1, d2 → d3): (a) images, (b) corresponding labels, and (c) segmentation results.

TABLE VI

ACCURACIES OF HY-MSDA AND THE SEGFORMER ON THE CLOUD DETECTION DATASETS. THE LARGER THE VALUE IS, THE BETTER THE PERFORMANCE OF THE METRIC

Method / Domain	Hy-MSDA Accuracy (%)	Segformer Accuracy (%)
d2→d1	89.7	95.0
d3→d1	90.4	
d2, d3→d1	94.3	
d1→d2	90.5	95.5
d3→d2	91.9	
d1, d3→d2	94.7	
d1→d3	91.5	96.2
d2→d3	91.9	
d1, d2→d3	96.0	

TABLE VII

PARAMETER COUNT, GFLOPS, MODEL SIZE, AND TRAINING TIME REQUIRED FOR DIFFERENT MODELS IN THE TASK OF RS SCENE CLASSIFICATION (s1, s2, s3 → s4). ↓ INDICATES THAT THE SMALLER THE METRIC VALUE IS, THE BETTER THE PERFORMANCE

Models	Parameters ($\times 10^6$) ↓	GFLOPs↓	Size (MB)↓	Training Time Cost (Hours)↓
M3SDA	33.1	34.2	167.5	10.3
MFSAN	33.0	34.8	170.0	10.2
LCt-MSDA	31.9	33.5	155.6	9.7
T-SVDNet	32.1	33.7	150.3	9.8
MCC-DA	29.8	31.3	140.8	8.9
PTMDA	30.5	32.5	131.1	14.5
SUMDA	31.1	35.6	160.5	14.0
RRL	30.8	32.8	146.5	8.2
Hy-MSDA (pure CNN)	30.3	31.5	120.7	7.5
Hy-MSDA (SA)	36.2	38.1	132.6	8.3
Hy-MSDA	29.5	30.2	113.8	7.8

data, as demonstrated in the figures, indicating the effectiveness of our approach in cloud detection tasks.

E. Analysis of Computational Metrics

To further evaluate the computational complexity, we conducted an analysis of different methods based on various metrics in Table VII. These metrics provide insights into the computational efficiency of each method during the training phase.

Table VII shows that Hy-MSDA outperforms the other methods in terms of the number of parameters, number of GFLOPs, model size, and training time cost (Hours). This indicates that Hy-MSDA has a more efficient and effective computational framework overall. In addition, we investigated the impact of hyperparameters during training on CNN backbones, CNN–Transformer backbones with standard self-attention (SA), and our proposed CNN–Transformer backbone using HW-SA, where the three settings denoted as Hy-MSDA (pure CNN), Hy-MSDA (SA), and Hy-MSDA in Table VII. The last three rows indicate significant increases in parameters, GFLOPs, model size, and training time when using the SA-based CNN–Transformer backbone, attributed to the computational demands introduced by the self-attention mechanism in Transformers. Conversely, employing our enhanced HW-SA approach resulted in reductions across these metrics. Although our method shows a slight increase in training time compared to using a pure CNN backbone, this marginal increase is overshadowed by improvements in parameters, GFLOPs, and model size metrics. Furthermore, considering the substantial accuracy gains achieved by the CNN–Transformer backbone, these minor increases in training time can be deemed negligible.

F. Analysis of Hyperparameters

Additionally, we assessed the range of values of the parameters α_2 and α_3 in (17), which are used to balance the consistency loss between the source and target domains as well as the consistency loss between the target domain and

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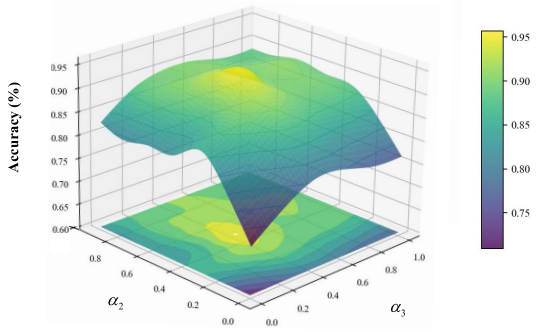


Fig. 14. Performance on different values of α_2 and α_3 on the RS scene classification dataset (s1, s2, s3 → s4).

TABLE VIII

CHOICE OF PARAMETERS. THE BOLD TEXT INDICATES THE BEST RESULT FOR THE CHOICE OF PARAMETERS

Parameters	Value	OA (%)	Precision (%)	Recall (%)	F1-Score (%)
Batch sizes	2	0.965	0.956	0.949	0.947
	4	0.971	0.969	0.958	0.962
	8	**0.977**	**0.982**	**0.973**	**0.966**
Learning rates	$2\times e^{-2}$	0.944	0.950	0.945	0.950
	$2\times e^{-3}$	0.958	0.969	0.951	0.959
	$2\times e^{-4}$	**0.977**	**0.982**	**0.973**	**0.966**

the generated target domain. Experiments were conducted on the RS scene classification dataset (s1, s2, s3 → s4) while varying the values of α_2 and α_3, with the resulting accuracy plots presented in Fig. 14.

It is noteworthy that when α_2 and α_3 approach 0, the model's performance deteriorates significantly. This observation indicates that both the consistency alignment loss between the source and target domains and the consistency loss between the target domain and the generated target domain are crucial for the model's effectiveness, thereby validating the importance of our consistency learning module. The optimal accuracy was achieved when α_2 was set to 0.45 and α_3 was set to 0.4. Consequently, we established α_2 at 0.45 and α_3 at 0.4 for our subsequent experiments.

Additionally, we investigated the fine-tuning of hyperparameters, focusing first on the batch size. Our experiments included testing batch sizes of 2, 4, and 8. The results indicated that a larger batch size generally improved the model's performance. Therefore, we selected a batch size of 8 as the optimal value for our final model.

Next, we examined various learning rates, specifically $2 \times e^{-4}$, $2 \times e^{-3}$, and $2 \times e^{-2}$. We observed that higher learning rates led to model instability and hindered convergence, whereas excessively low learning rates resulted in prolonged training times. On the basis of these observations, we determined that a learning rate of $2 \times e^{-4}$ is the most effective. The results of these experiments are summarized in Table VIII.

V. CONCLUSION

The proposed Hy-MSDA shows effectiveness in MUDA tasks through the integration of a hybrid CNN–Transformer backbone, a consistency learning module, and a dynamic weighting strategy. The hybrid CNN–Transformer backbone enables Hy-MSDA to capture both local details and long-range dependencies. The consistency learning module maintains domain-invariant features, enhancing performance with global high-level features and rich semantic information. Additionally, the dynamic weighting strategy considers the data distributions and the model's learning capability, effectively balancing information from multiple sources. All the aforementioned components are closely interconnected and mutually reinforce each other to achieve collaborative cross-domain adaptation. Extensive experimental results validate the effectiveness of the proposed network in both scene classification and landcover segmentation tasks.

In this work, we conducted scene classification and landcover segmentation experiments on the shared categories across all the domains. However, it is important to acknowledge that in practical scenarios, the target domain may contain categories that are not present in the source domain label sets. These variations pose challenges for the generalization and application of MUDA methods. To address these challenges, our future work will focus on developing MUDA, which can handle inconsistent categories and extend the applicability of the model to target domains with categories not seen in the source domains.

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